1. Light

Introduction

The purpose of industrial lighting is to provide an efficient and comfortable seeing of industrial tasks and to help provide a safe working environment. Adequate lighting is not for safety alone. Lighting adequate for seeing production and inspection tasks will be more than is needed for safety. Light is more for comfort and convenience than for safety.

It has been shown that adequate industrial lighting results in many benefits; for example:

1. Promotes reduced production and inspection mistakes.
2. Increases production.
3. Reduces accidents.
4. Improves morale.
5. Improves housekeeping.

What Is Light?

The nature of light is not easy to understand. The question, "What is light?" has been extremely elusive throughout the history of science.

Near the end of the seventeenth century, there were two theories to explain the nature of light; the particle or corpuscular theory and the wave theory. In the nineteenth century, the discovery of interference and diffraction--the bending of light as it passes through different media--made the wave theory of light the predominant theory; i.e., interference and diffraction could not be explained adequately by the particle or corpuscular theory.

In the late 1800's, light was thought to be electromagnetic waves, which at certain frequencies could be seen by the human eye. This conceptualization of light was primarily used to explain the propagation of light. To conceptualize the propagation of light, an understanding of electrical forces and fields and magnetic forces and fields is needed.

To illustrate electrical forces, the simple case of an electrical charge at rest will be discussed. Each atom has a positively charged core, the nucleus, which is surrounded by negatively charged electrons. The nucleus consists of a number of protons, each with a single unit of positive charge and one or more neutrons (except for hydrogen). A neutron is a neutral

particle. Normally, an atom is in a neutral or uncharged state because it contains the same number of protons as electrons. If, for some reason, a neutral atom loses one or more of its electrons, the atom will have a net positive charge and is referred to as an "ion." A negative ion is an atom which has gained one or more additional electrons. That is, an object which has an excess of electrons is negatively charged; and an object which has a deficiency of electrons is positively charged. Objects can be electrically charged in many different ways.

It is known that objects with the same charge repel each other, while objects with opposite charges attract each other. That is, some force exists between charged objects. In 1784 Charles Augustine de Coulomb found that the force of attraction or repulsion between two charged objects is inversely proportional to the square of the distance separating them. Coulomb's law can be stated as:

> The force of attraction or repulsion between two point charges is directly proportional to the product of the two charges and inversely proportional to the square of the distance between them.

From Coulomb's law, the following may be written:

$$F \propto \frac{qq'}{r^2}$$

or

$$F = \frac{kqq'}{r^2}$$

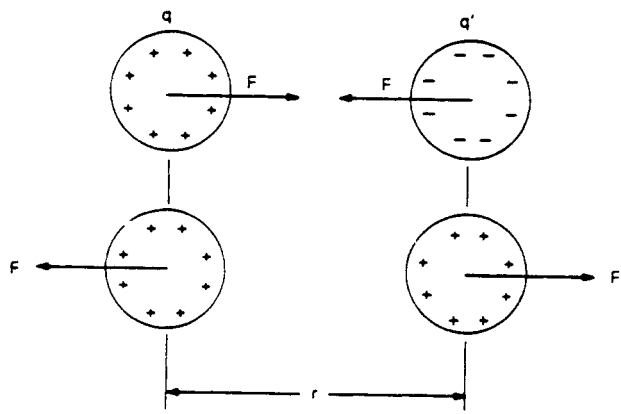
where F denotes the magnitude of force; q and q' represent the magnitude of two charges; r represents the distance between the charges; and k represents Coulomb's constant, a proportionality constant that takes into account the property of the medium separating the charged bodies. The units attached to q, q', and k are of no concern for conceptual development. This relationship can be graphically illustrated by Figure 5.1.1.

Electrically charged bodies then exhibit a force. The presence of an electrically charged object alters the space around it. This alteration in the surrounding space can be described by introducing the concept of fields. An electrical field is said to exist in a region of space in which an electric charge will experience an electrical force. The strength of the electrical field at any point will be proportional to the force a given charge experiences at any point; i.e., the strength of an electrical field can be represented by the force per unit charge. The electrical field intensity, E, is then defined at a point in terms of the force, F, experienced by an arbitrary positive charge, +q, when it is placed at that point. Thus

$$E = \frac{F}{+q}$$

Figure 5.1.1

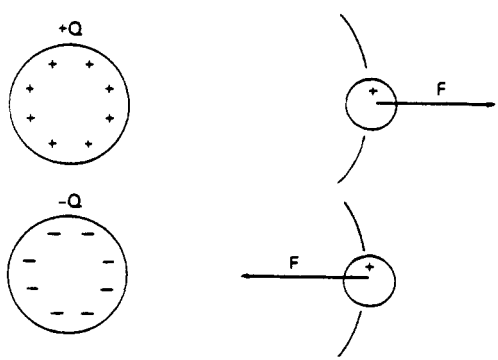
Illustration of Coulomb's law.



Since the electrical field intensity is defined in terms of a positive charge, its direction at any point would be the same as the electrostatic (at rest) force on a positive charge at that point. (See Figure 5.1.2.)

Figure 5.1.2

Direction of electrical fields.



On this basis, the electrical field in the vicinity of a positive charge would be outward or away from the charge, while in the vicinity of a negative charge the direction of the field would be inward or toward the charge. (See Figure 5.1.3.)

Figure 5.1.3

Direction of electrical fields.



If an electrically charged particle is brought into the field created by another electrically charged particle, then

$$E = \frac{F}{q'} = \frac{kqq'/r^2}{q'} = \frac{kq}{r^2}$$

where E denotes the electrical field intensity, F denotes force, r denotes the distance between the two emitted charges, q and q' denote the magnitude of the charges, and k denotes Coulomb's constant.

Magnetic forces and fields are similar to electrical forces and fields. The magnetic law of forces states: Like magnetic poles repel each other while unlike magnetic poles attract each other. In the eighteenth century, de Coulomb discovered that (1) the force of attraction or repulsion between the poles of two magnets is inversely proportional to the square of the distance, r, between the poles; and (2) the force of attraction or repulsion between two poles is along a line joining the two poles and directly proportional to the product of the pole strengths, p_1 and p_2. These two statements may be combined to form a mathematical statement:

$$F = \frac{kp_1p_2}{r^2}$$

where F denotes the force between two poles of strength p_1 and p_2 which are separated by a distance, r. The value of the proportional constant, k, depends upon the units chosen and the medium surrounding the magnets.

Every magnet is surrounded by a space in which the magnetic effects are present. These regions are called "magnetic fields." The strength of the magnetic field at any point is referred to as the magnetic field intensity, H, and is defined in terms of the force exerted on a unit north pole; i.e. the magnetic field intensity, H, at any point is the magnetic force per unit north pole placed at that point:

$$H = \frac{F}{p}$$

where H denotes the magnetic field intensity, F denotes force, and p denotes the magnitude of the unit north pole.

A more useful expression for computing the magnetic field intensity is given by

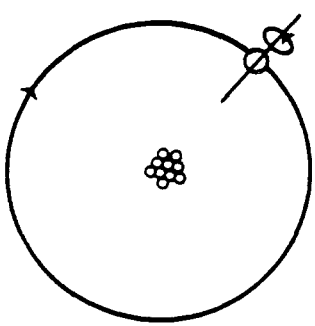
$$H = \frac{F}{p'} = \frac{kpp'/r^2}{p'} = \frac{kp}{r^2}$$

where p' denotes a test pole placed distance r from the pole p, and the remaining symbols are defined as before.

Magnetism is believed to result from movements of electrons within the atoms of substances. That is, magnetism results from a change in motion. The magnetic polarity of two atoms stems primarily from the spin of electrons about their own axis and is due also to their orbital motions around the nucleus. (See Figure 5.1.4.) As can be seen, magnetism is closely related to electrical phenomena.

Figure 5.1.4

A charge in motion.



If compass needles were placed around an electrical current, a magnetic field would be produced. (Figure 5.1.5.) If a moving charged particle creates a magnetic field, will a moving magnetic field create an electrical field? The answer is "yes." (Figure 5.1.6) This figure consists of a wire with some loops in it and a regular bar magnet. If the magnet is moved up and down, the meter to the right will register an electrical current or electrical field. A magnetic field is created by the movement of a charged particle, and the movement of a magnetic field creates an electrical field.

The upper part of Figure 5.1.7 indicates that when a wire with no initial current is moved downward, the charges in the wire experience a deflecting force perpendicular to their motion. Since there is a conducting path made by the wire in this direction, the electrons follow it, thereby constituting a

Figure 5.1.5

A moving charge creates a magnetic field.

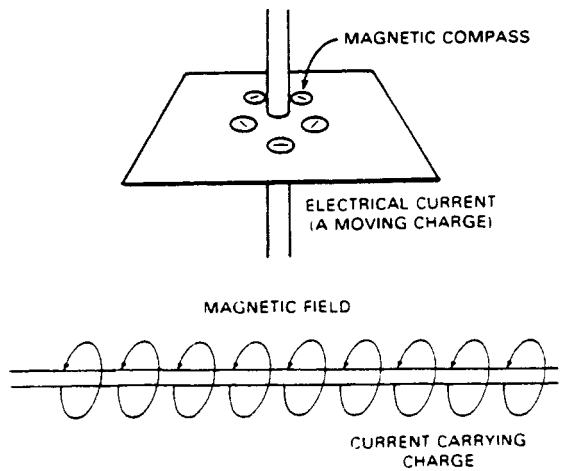


Figure 5.1.6.

A moving magnetic field creates an electrical field.

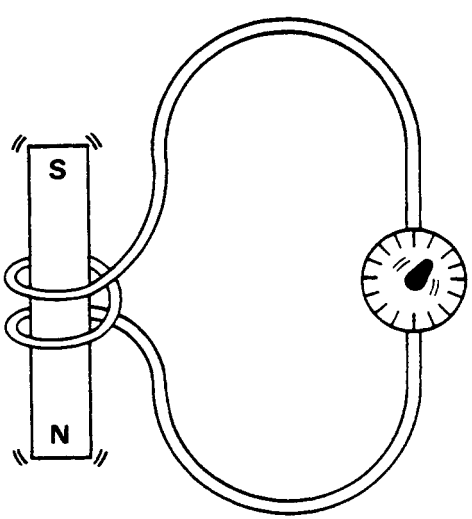
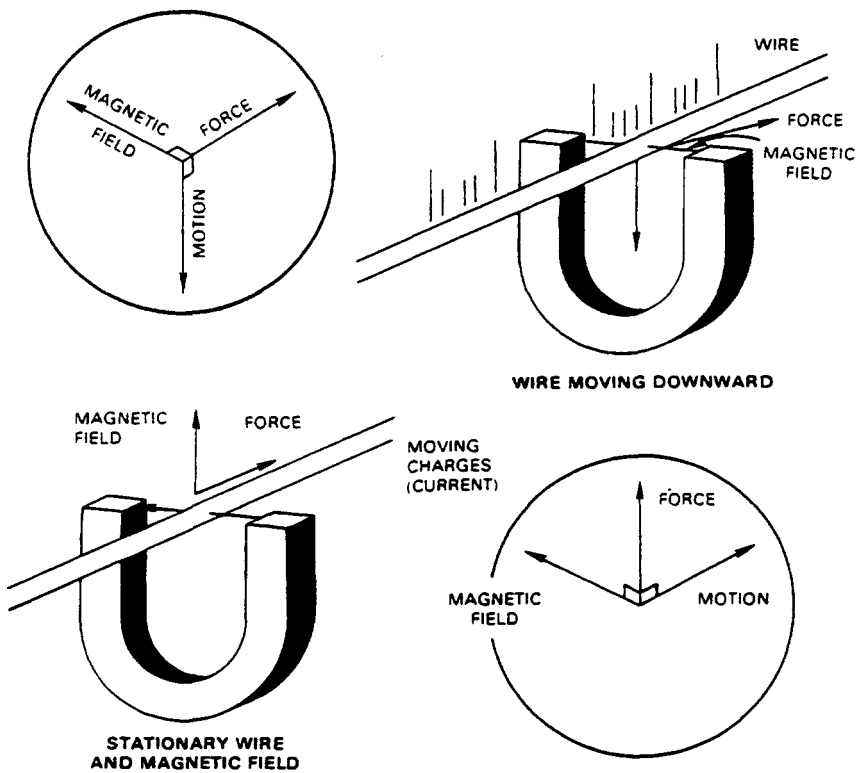


Figure 5.1.7

Moving charges experience a force that is perpendicular to the magnetic field lines they traverse.



current. In the lower half of the diagram where the magnet is stationary and the wire is stationary, when a current moves through the wire to the right, there is a perpendicular upward force on the electron. Since there is no conducting path upward, the wire is tugged upward along with the electrons of the charged particles. The relationship between magnetic fields and electrical fields can be summarized by saying that moving charges experience a force that is perpendicular to the magnetic field lines that they traverse.

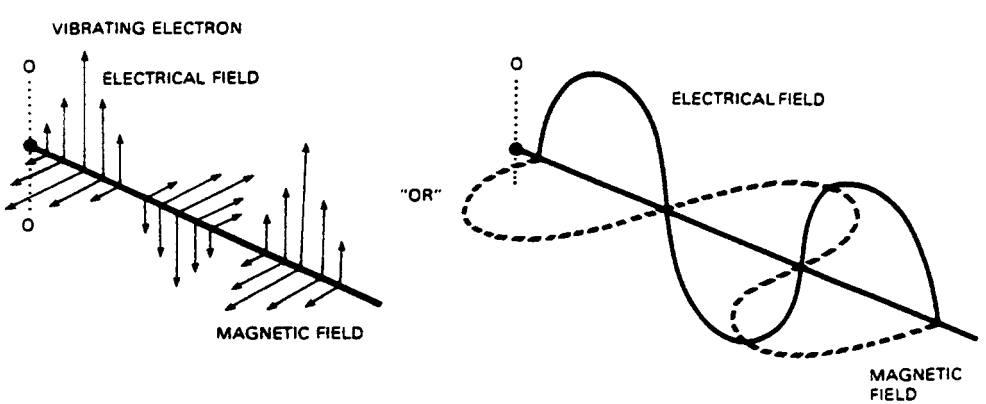
The fact that a magnetic field induces an electrical field and a moving electrical field produces a magnetic field is how an electromagnetic wave is produced. Consider an electrical charge vibrating back and forth at a certain frequency. The charge, because it is a moving charged particle, will produce an electrical field around it. The moving charge, if it vibrates back and forth, creates a magnetic field. However, the magnetic field is also changing or moving; and the moving magnetic field produces an electrical field. Thus, the two fields are mutually induced. The changing magnetic field induces an

electrical field which induces a magnetic field, etc., and electromagnetic waves are produced.

Consider first the initial magnetic field induced by the moving charge. This changing magnetic field induces a changing electrical field, which in turn induces a magnetic field. The magnitude of this further induced magnetic field depends not only on the vibrational rate of the electrical field but also on the motion of the electrical field or the speed at which the induced field emanates from the vibrating charge. The higher the speed, the greater the magnetic field it induces. At low speeds, electromagnetic regeneration would be short lived because the slow-moving electrical field would induce a weak magnetic field which in turn would induce a weaker electrical field. The induced fields become successively weaker, causing the mutual induction to die out. But what about the energy in such a case? The fields contain energy given to them by the vibrating charge. If the fields disappear with no means of transferring energy to some other form, energy would be destroyed. Low-speed emanation of electrical and magnetic fields is incompatible with the law of conservation of energy. At emanating speeds too high, on the other hand, the fields would be induced to ever-increasing magnitudes with a crescendo of ever-increasing energies--again clearly in contradiction with the conservation of energy. At some critical speed, however, mutual induction would continue indefinitely with neither a loss nor a gain in energy. This critical speed without loss or gain of energy is 186,000 miles per second--the speed of light. Thus, energy in an electromagnetic wave is equally divided between electrical and magnetic fields that are perpendicular. Both fields oscillate perpendicular to the direction of the wave propagation. (Figure 5.1.8)

Figure 5.1.8

Representation of electromagnetic wave.



Electromagnetic Spectrum

In 1885 H. R. Hertz showed that radiation of electromagnetic energy can occur at any frequency. All electromagnetic waves travel at the same speed in a vacuum. The waves are different from one another in their frequency and wavelength. The relationship between velocity, frequency, and wavelength is as follows:

$$C = f\lambda$$

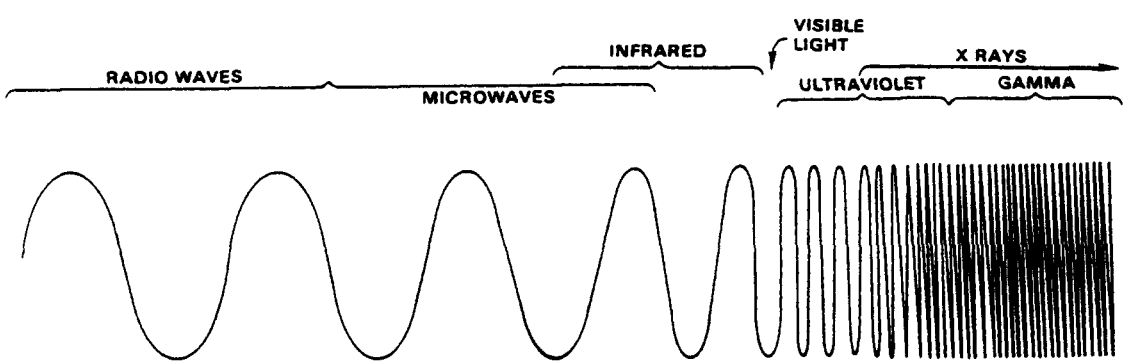
where C denotes velocity, f denotes frequency, and λ denotes wavelength.

If the speed or velocity of electromagnetic waves is constant, then when the frequency changes, the wavelength must change. The higher the frequency of the vibrating charge, the shorter the wavelength.

Figure 5.1.9 shows the electromagnetic spectrum. It extends from radio waves to gamma waves. In all sections of the electromagnetic spectrum, the waves are the same in nature; they differ only in frequency and wavelength.

Figure 5.1.9

Electromagnetic spectrum.



Electromagnetic waves in principle can have any frequency from zero to infinity. The classification of electromagnetic waves according to frequency is called the "electromagnetic spectrum." Electromagnetic waves with frequencies of the order of several thousand hertz (kilocycles/sec) are classified as radio waves. The VHF (very high frequency) television band starts at about 15 million hertz (megacycles/sec). Still higher frequencies are called "microwaves" followed by infrared waves often called "heat waves." Further still is visible light which makes up only one percent of the measured electromagnetic spectrum. Beyond light, the higher frequencies extend into the ultraviolet, X-ray, and gamma-ray regions. There is no sharp distinction

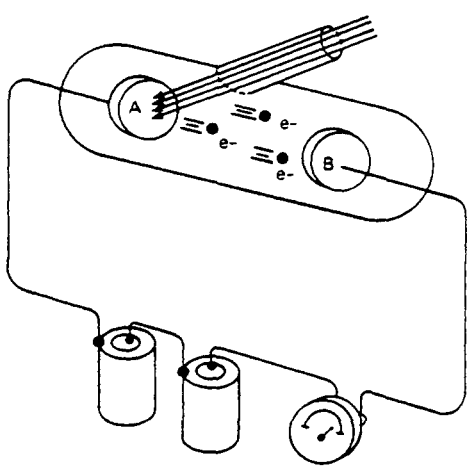
between these regions which actually overlap each other. The spectrum is simply broken up into these arbitrary regions for classification.

The Quantum Theory

Around 1885 light was thought to be wave-like and not particle-like. However, in 1887 Hertz noticed that an electrical spark would jump more readily between charged fields when their surfaces were illuminated by the light from another spark. This observation is commonly known as the "photoelectric effect." The arrangement for the photoelectric effect can be seen in Figure 5.1.10.

Figure 5.1.10

Photoelectric effect.



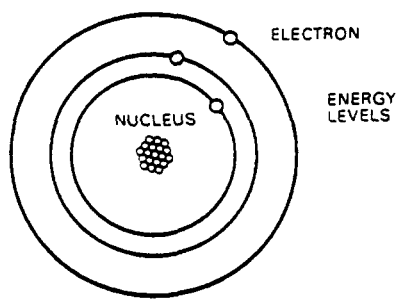
Light shining on the negatively charged photosensitive metal plate liberates electrons, which are attracted to the positive plate, producing a measurable current. This photoelectric effect could not be explained by the electromagnetic wave theory of light. The brightness of light in no way affected the energies of the ejected electrons. If light was accepted to be electromagnetic radiation, then the stronger electric fields of bright light would surely interact with electrons, causing them to eject at greater speeds and, thus, greater energies. Yet, this was not the case. No increase in electron kinetic energy was detected. A weak beam of ultraviolet light produced a given number of electrons but much higher kinetic energies. This was puzzling; the wave theory of light could not explain this phenomenon.

In an attempt to bring experimental observations into agreement with theory, Max Planck published his quantum hypothesis. He found that the

problem with the electromagnetic theory of light lay with the assumption that energy is radiated continuously. He postulated that electromagnetic energy is absorbed or emitted in discrete packets or quanta, referred to as photons. Planck postulated that the energy states of electrons in atoms are quantized; i.e., electrons can only vibrate with certain discrete amounts of energy. An electron farther away from the nucleus has a greater potential energy with respect to the nucleus than the electron nearer the nucleus; the furthermost electron is at a higher energy level. When an electron in an atom is raised to a higher energy level, the atom is said to be excited. The higher level of the electron is only momentary. The electron loses its temporarily acquired energy when returning to a lower energy level. (Figure 5.1.11)

Figure 5.1.11

Energy levels.



Radiation occurs when an electron makes the transition from a higher energy state to a lower energy state; i.e., when the atom becomes de-excited. This energy is in quanta. The electron moves in discrete steps from higher levels to lower levels. Each element has its own number of electrons; each element also has its own characteristics of energy levels. An electron's dropping from a higher energy level to a lower energy level in an excited atom emits energy in photons or quanta with each jump. So for each element, the amount of energy will be different.

Planck postulated that the energy of the resulting quanta of radiation would be equal to the difference in the energy state of the atom. Further, he postulated that the frequency of the emitted radiation is proportional to this energy difference. Planck's equation can be written as

$$E = hf$$

where E denotes the energy of the photon, f denotes the frequency of radiation, and h is the proportionality factor called Planck's constant.

Thus, a photon or quantum of infrared radiation has a very small energy; a quantum of green light, a small energy; and a quantum of ultraviolet light, a

larger energy. The greater the radiation frequency of the quantum, the greater the energy.

With Planck's theory, the nature of light was seen to be dualistic--it had wave-like properties and particle properties (quanta or photons being released upon de-excitation). It is customary to discuss the wave-like properties of light when the propagation of light is discussed, and it is customary to use the particle theory when the interaction of light with matter is discussed. Light may be thought of as radiant energy transported in photons which are carried along by a wave field.

The Emission Spectra

Every element has its own characteristic pattern of electron levels. An electron dropping from higher to lower energy levels in an excited atom emits a photon with each jump. Many frequency characteristics of an atom are emitted corresponding to the many paths the electron may take when jumping from level to level. These frequencies combine to give light from each excited atom its own characteristic color. This unique pattern can be seen when the light is sent through a prism.

Each component color is focused at a definite position according to its frequency and form. If the light given off by a sodium vapor lamp is analyzed, a single yellow line is produced. If the width of the ray of yellow light could be narrowed, it would be found that the line is composed of two very close lines. These lines correspond to the two predominant frequencies of light emitted by the excited sodium atoms. The rest of the spectrum is dark. (There are many other lines too dim to be seen with the naked eye.)

This situation is not unique in sodium. Examining the light from a mercury vapor lamp reveals two strong yellow lines close together (but in different positions than those of sodium), a very intense green line, and several blue and violet lines. Similar but more complicated patterns of lines are found in light emitted by a neon tube. The light emitted by every element in a vapor state produces its own characteristic pattern of lines. These lines correspond to the electron transitions between atomic energy levels and are characteristic of each element as are the fingerprints of people.

Incandescence

Light emitted from a neon tube is red because the average difference in neon energy level is proportional to the frequency of red light. Light emitted by a common incandescent lamp, however, is white. All frequencies of visible radiation are emitted. Does this mean that tungsten atoms making up the lamp filament are characterized by an infinite number of energy levels? The answer is a definite "no." If the filament were vaporized and then excited, the tungsten gas would emit a finite number of frequencies, producing an overall bluish color. The frequency of light emitted by atoms depends not only upon the energy levels within the atom but also on the spacings between neighboring atoms themselves. In a gas, the atoms are far apart. Electrons

undergo transition between energy levels within the atom quite unaffected by the presence of neighboring atoms. But when the atoms are closely packed, as in a solid, the electrons of the outer orbits make transitions not only within the energy levels of the parent atoms but also between the levels of neighboring atoms. These energy level transitions are no longer well defined but are altered by interactions between neighboring atoms, resulting in an infinite variety of energy level differences, thence, an infinite number of radiation frequencies. And this is why tungsten filament light is white.

Mercury vapor light is bright and less expensive than incandescent lamps. Most of the energy in incandescent lamps is converted to heat, while most of the energy put into mercury vapor lamps is converted to light. As the filament in a tungsten filament lamp becomes heated, wider energy level transitions take place, and higher frequencies of radiation are emitted. A hotter filament produces a whiter light.

Fluorescence and Fluorescent Lamps

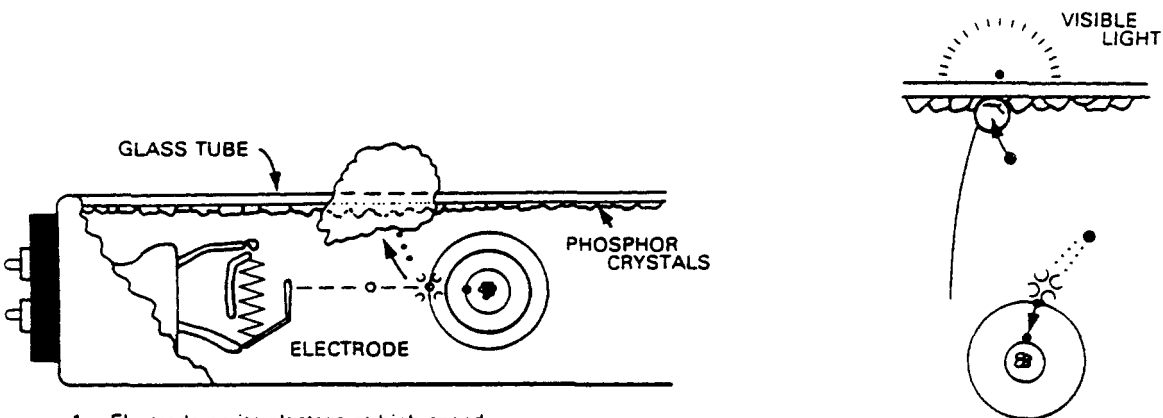
Atoms absorb light as well as emit light. An atom will most strongly absorb light having the same frequency or frequencies to which it is tuned, the same frequency it emits. For example, when a beam of white light passes through a gas, the atoms of the gas absorb selected frequencies. This absorbed energy is reradiated in all directions instead of in the directions of the incident light.

Some atoms become excited when absorbing a photon of light. Ultraviolet light has more energy per photon than lower frequency light. Many substances undergo excitation when illuminated by ultraviolet light. When a substance excited by ultraviolet light emits visible light upon de-excitation, this action is called fluorescence. What happens in some of these materials is that a photon of ultraviolet light collides with an atom of the material and gives up its energy in two parts. Part of the energy goes into heat, increasing the kinetic energy of the entire atom. The other part of the energy goes into excitation, boosting the electron to a higher orbit. Upon de-excitation, this part of the energy is released as a photon of light. Since some of the energy of the ultraviolet photon is converted to heat, the photon emitted has less energy and, therefore, lower frequency than the ultraviolet photon. That is, the secondary photon of light that is released is of less energy than the primary photon since some energy goes to heat; thus, it is of a lower frequency. Light emitted from fluorescent lamps is produced by primary and secondary excitation processes. The primary process is excitation of a gas by electron bombardment; and the secondary process is excitation by ultraviolet photons, fluorescence. The common fluorescent lamp consists of a cylindrical gas tube with electrodes at each end. (See Figure 5.1.12.) As in the neon sign tube, electrons are boiled off from the electrodes and forced to vibrate back and forth at high speeds within the tube by an AC voltage; and the tube is filled with very low-pressure mercury vapor which is excited by the impact of high-speed electrons. As the energy levels in the mercury are relatively far apart, the resulting emission of light is of very high frequency, mainly ultraviolet light. This is the primary excitation process. The secondary process occurs when the ultraviolet light impinges

upon a thin coating of powdery material made up of phosphors on the inner surface of the glass tube. The phosphors are excited by the absorption of the ultraviolet photon and give off a multitude of lower frequencies in all directions that combine to produce white light. Different phosphors can be used to produce different colored lights.

Figure 5.1.12

Fluorescent lamp.



1. Electrode emits electron at high speed.
2. Collides with atom (usually mercury atom).
3. Collision causes excitation. Upon de-excitation, a photon is released (ultraviolet light).
4. Ultraviolet photon hits phosphor crystals where *excitation takes place.*
5. Upon de-excitation, a photon is released, producing visible light.

Comparing Different Light Sources

The process of excitation and de-excitation (and the release of photons) explains how light is emitted. Mercury vapor lamps, fluorescent lamps, and incandescent lamps all work on the same principle.

Incandescent lamps produce extremely white light. The hotter the filament, the whiter the light. However, the hotter they burn, the weaker they get and the more wear and tear on the filament, thus decreasing the life of the lamp.

High-pressure mercury lamps are about twice as efficient as filament lamps because less energy is converted into heat. One of the disadvantages of high-pressure mercury lamps is the delay in starting and restarting them. Several minutes are required for the lamps to reach full brightness. In cases of power interruption, the lamps will not restart until the arc tube has cooled sufficiently for the mercury vapor to condense (about five minutes). This disadvantage can be overcome by installing a filament lamp along with a mercury lamp.

Fluorescent lamps (low-pressure mercury lamps) are about three times as efficient as filament lamps. A coating on the glass filters out the radiation that would be harmful to the eyes or to the skin. In the past, fluorescent lamps were started with a starter that heated the electrodes at the end of the tube. Modern lamps are started with a ballast; they have sufficient voltage to start the lamp immediately. Fluorescent bulbs give long life, about 7500 hours. The lamp is affected by the number of starts. Lamps will last longer if started less frequently.

When deciding what type of lamp to use, one has to look at both efficiency and cost. Where costs are low, a shorter lamp life would be sufficient. Where lamps are costly or the labor costs of replacing them are high, a longer-life lamp is more economical.

Incandescent filament lamps come in various shapes and sizes. The lamp bulbs are designated by a letter code followed by a numeral. The letter indicates the shape (straight, S; flame, F; globe, G; general service, A; tubular, T; pear shape, PS; parabolic, PAR; and reflector, R). The number indicates the size--the diameter of the bulb in eighths of an inch. Thus, a T-12 lamp is a tubular lamp that is 12/8 inches or 1.5 inches in diameter. Incandescent lamps also come with different kinds of bases: disc, candelabra, intermediate, mogul, bayonet, bipost, etc.

Mercury vapor lamps are designated by ASA nomenclature; e.g., H33-1-CL/C, where H denotes mercury; 33-1, the ballast number; CL, arbitrary letters designating physical characteristics of the lamp such as bulb size, shape, material, and finish; and C indicates the color of the light.